

AERATED DRILLING

Fundamentals, hydraulics



One decision changes the whole circulating system

Add gas



**Lower average
density**



**New risks
and controls**

Aerated drilling is not simply “mud plus air.”

Density, compressibility, transport, pressure response, surface handling and well control all change together.

By the end, you can explain—and challenge—the plan

01

Explain

Describe why gas is added and how it changes bottomhole pressure.

02

Trace

Follow liquid, gas and returns through the complete equipment path.

03

Interpret

Read the operating window using pressure, transport and equipment limits.

04

Respond

Choose safe actions during start-up, connections, losses and abnormal returns.

Four modules build from concept to field response

MODULE	SUBJECT	CLASSROOM QUESTION	TIME
01	Principles	What aeration changes	45 min
02	Equipment	How the system is connected	45 min
03	Hydraulics	How the operating window is built	50 min
04	Operations	How the team starts, controls and responds	50 min

Two 10-minute breaks · exercises are included in each module

Local gas fraction and phase continuity define the drilling regime

Lower local gas fraction

Higher local gas fraction



Aerated liquid

Continuous liquid phase
Gas dispersed as bubbles

Stable foam

Connected liquid films
Structured carrying phase

Mist / unstable foam

Gas-continuous phase
Liquid dispersed as droplets

Dry air / gas

No injected liquid
Lowest mixture density

The target is pressure control inside a narrow window

Fracture / loss limit

Too high: losses, formation damage, induced fractures

OPERATING WINDOW

Enough pressure for stability and control—low enough for the drilling objective

Too low: influx, instability, excessive flow or barrier challenge

Pore / formation pressure

Principles of aerated drilling

Understand what changes when compressible gas enters a liquid circulation system.

45 minutes · concept + exercise

Aerated liquid keeps a continuous liquid phase

Liquid carries

- cuttings and fines
- heat away from bit and BHA
- chemicals for foam, corrosion and fluid properties

Gas changes

- average mixture density
- annular velocity and flow regime
- pressure response and compressibility
- surface return volume

Key distinction: the gas fraction changes continuously with pressure and temperature.

Aeration is selected for a specific drilling objective

01

Reduce pressure

Lower the average hydrostatic contribution and potentially reduce losses or reservoir invasion.

02

Improve performance

Reduce chip hold-down and increase ROP when the formation and hole condition are suitable.

03

Protect the objective

Preserve permeability, manage depleted or fractured intervals, and keep returns manageable.

Selection still depends on stability, influx potential, water production, equipment, discharge controls and barrier philosophy.

Gas expands toward surface—and the return rate accelerates

NEAR SURFACE

**Large actual gas volume
and fastest return flow**

MID-DEPTH

**Bubbles expand as
pressure falls**

DEEP

**High pressure keeps
gas volume small**



Design surface equipment for actual return volume—not only the standard injection rate.

Hole cleaning is a race between upward flow and particle slip



A practical transport check

Transport ratio $\approx$

$$1 - \text{slip velocity} \div \text{upward mixture velocity}$$

- positive upward margin is required
- larger or heavier cuttings slip faster
- washouts reduce local velocity
- connections remove circulation support

Plain language: the upward conveyor must beat the cutting's tendency to fall.

Foam quality is gas volume fraction at local pressure and temperature

Q_f

Definition

$$Q_f = V_g / (V_g + V_l)$$

Use gas and liquid volumes at the same local P–T. Convert standard gas volume to downhole conditions before calculating quality.

≈52–95%*

Common teaching band

Useful classroom range—not a universal boundary.

Below ~52%: bubbly / aerated-liquid behavior.

Near ~95–98% in some systems: films thin; rupture can produce gas-continuous mist.

↑ Q_f

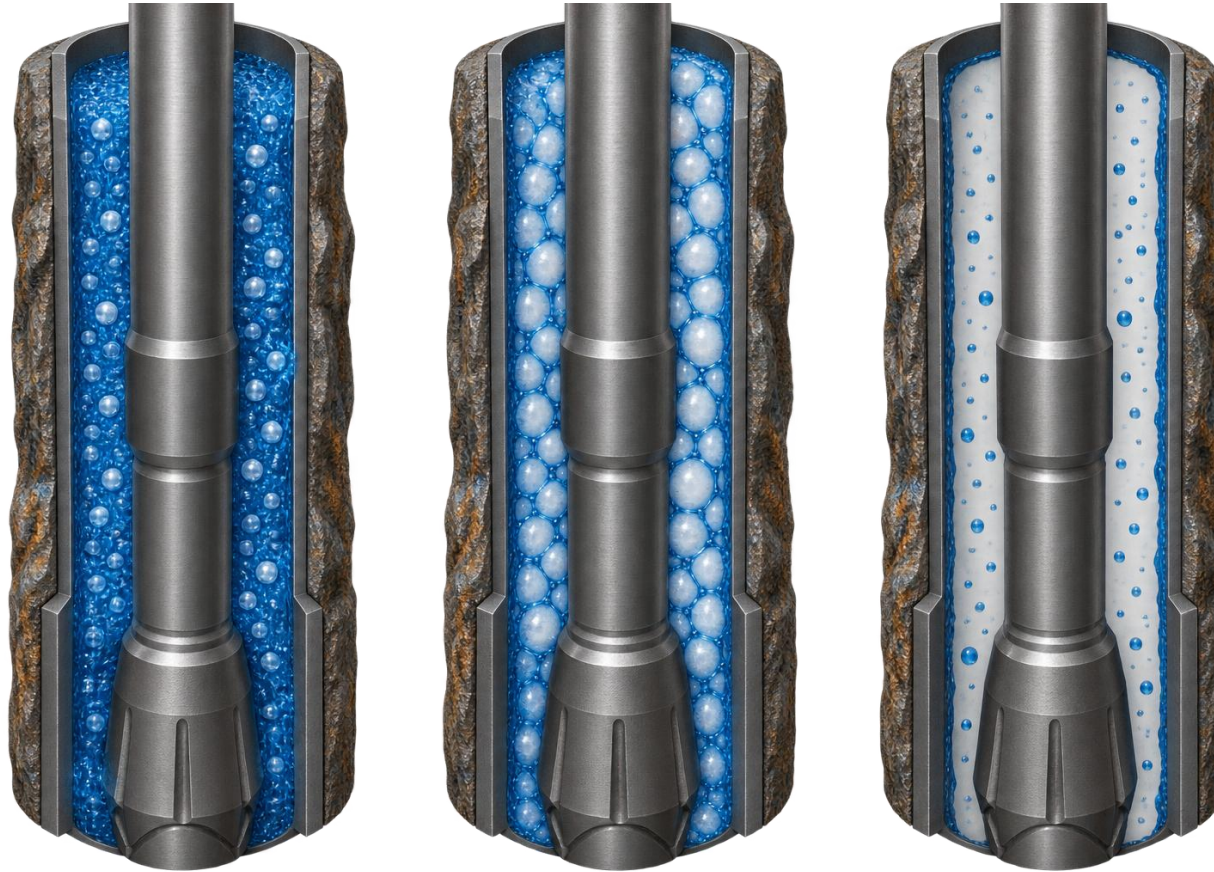
Field trade-off

Raising Q_f lowers mixture density, but stability can weaken near the dry limit.

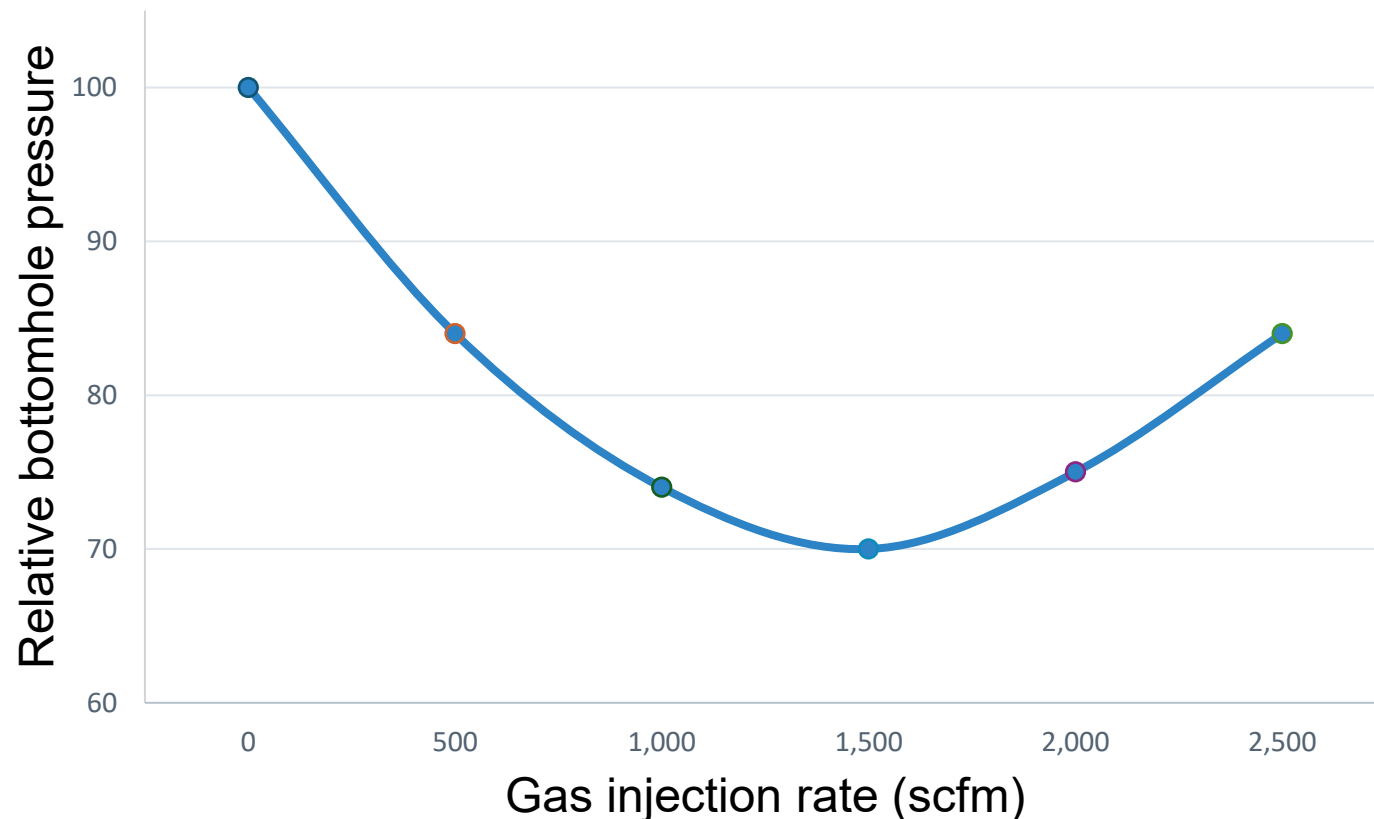
Verify with formulation-specific tests at representative pressure, temperature, salinity and shear.

Foam quality ≠ foamer dose: 70% quality means 70% gas by local volume; foamer concentration is chemical % of the liquid.

Increasing gas fraction: aerated liquid → stable foam → mist



More gas does not always mean lower bottomhole pressure



Density effect

Initially, more gas lowers average density and hydrostatic pressure.

Friction effect

At higher rates, friction and acceleration can reverse the trend.

Illustrative only—actual shape requires a validated model.

Exercise: choose a fluid mode before choosing a rate

Scenario

- fractured geothermal interval
- total losses with conventional mud
- possible hot-water inflow
- surface separator available
- air capacity is limited

Discuss in groups

1. What is the primary objective?
2. Which fluid mode would you screen first?
3. What could make that choice unsafe or ineffective?
4. What data must be confirmed before rates are set?

There is no single correct rate—the quality of assumptions matters.

Module 1: keep three relationships in view

01

Gas ↔ density

Gas reduces average density, but its downhole volume changes with pressure and temperature.

02

Velocity ↔ transport

Cleaning depends on upward mixture velocity relative to particle slip and local hole geometry.

03

Rate ↔ friction

Higher gas rate can eventually increase friction and bottomhole pressure.

10

minute break

Return ready to connect the equipment to the hydraulics.

Equipment and flow path

02

Trace energy, fluid and returns from the compressor to the separator—and identify every control point.

45 minutes · system walk-through +
exercise

The system is one pressure-connected chain



A restriction, leak or capacity limit anywhere can change the pressure response everywhere.

Compressor capacity is defined by both flow and pressure



Selection checks

- required standard gas rate
- pressure at the injection point
- altitude and temperature derating
- redundancy and turndown
- fuel, cooling and condensate handling

FLOW

Primary compressor supplies volume

PRESSURE

Booster raises discharge pressure

MATCH BOTH

Nameplate scfm alone is not capacity

Air and nitrogen solve different parts of the risk picture

CONSIDERATION	COMPRESSED AIR	MEMBRANE-GENERATED N ₂
Combustion support	Contains oxygen	Oxygen reduced below operating target
Package complexity	Simpler train	Membrane unit + matched compression
Delivered flow	Generally higher for same inlet package	Product flow reduced by separation process
Corrosion tendency	Oxygen contributes to corrosion	Lower oxygen; CO ₂ /H ₂ S/water still matter
Selection basis	Formation and fire risk acceptable	Hydrocarbon / combustion risk requires inerting

The injection point changes hydraulics and failure modes

A

Standpipe injection

Gas enters upstream of the drillstring. Simple surface arrangement, but all downstream components see the multiphase mixture.

B

Parasite string

A separate line injects gas at depth. It changes the pressure profile and requires dedicated integrity and placement controls.

C

Concentric / special string

Dedicated flow path can support specific operations, with added equipment and connection complexity.

The drilling program must define check valves, isolation, non-return protection and the response to backflow.

The rotating control device manages returns—it is not the only barrier

RCD cutaway



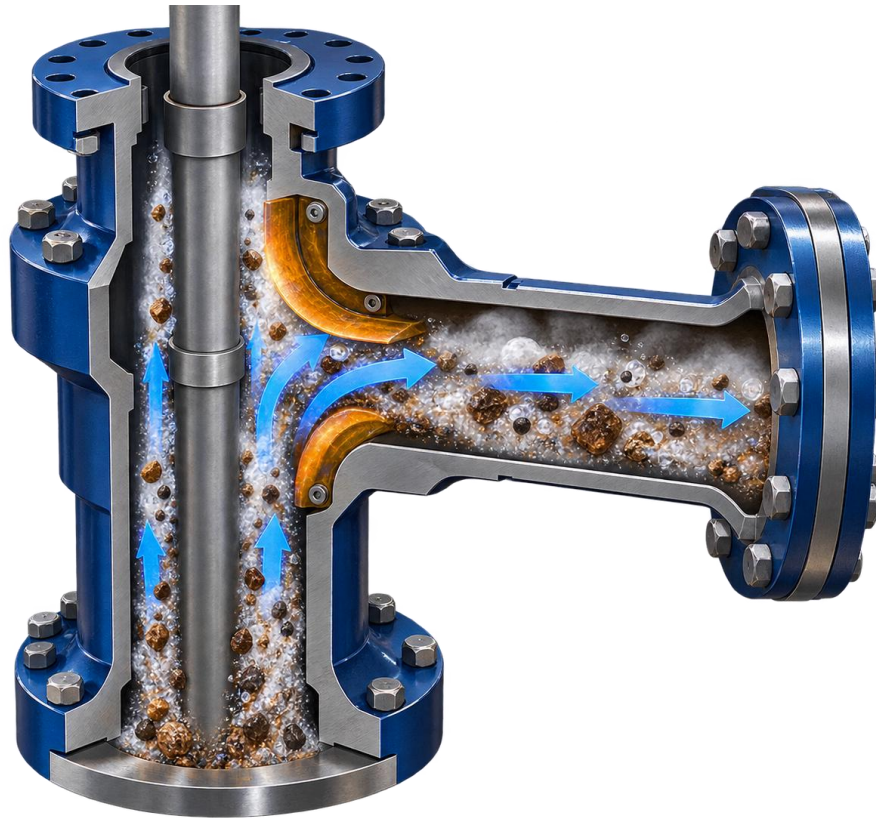
Rotating seal · bearings · side outlet

Barrier system still includes

- BOP stack and tested wellhead components
- drillstring non-return / inside safety devices
- choke and isolation valves
- documented pressure ratings—static and dynamic
- procedures for stripping, connections and shutdown

Never use a component outside the weakest applicable rating.

Banjo Box redirects high-energy geothermal returns



1 RECEIVE UPFLOW

Steam/air, liquid and cuttings rise in the annulus.

2 TURN INTO BLOWIE LINE

The heavy-walled tee redirects the stream about 90° away from the wellhead.

3 ABSORB EROSION

Inspect, rotate or replace the sacrificial insert before wall loss.

A Banjo Box redirects flow—it does not separate phases or replace the RCD/BOP barrier system.

Good control starts with synchronized measurements

01

Injection

- gas scfm
- liquid gpm
- injection pressure
- chemical rates

02

Rig

- SPP
- torque / rpm
- WOB / ROP
- pump strokes

03

Returns

- surface pressure
- flow / temperature
- gas detection
- pit volume

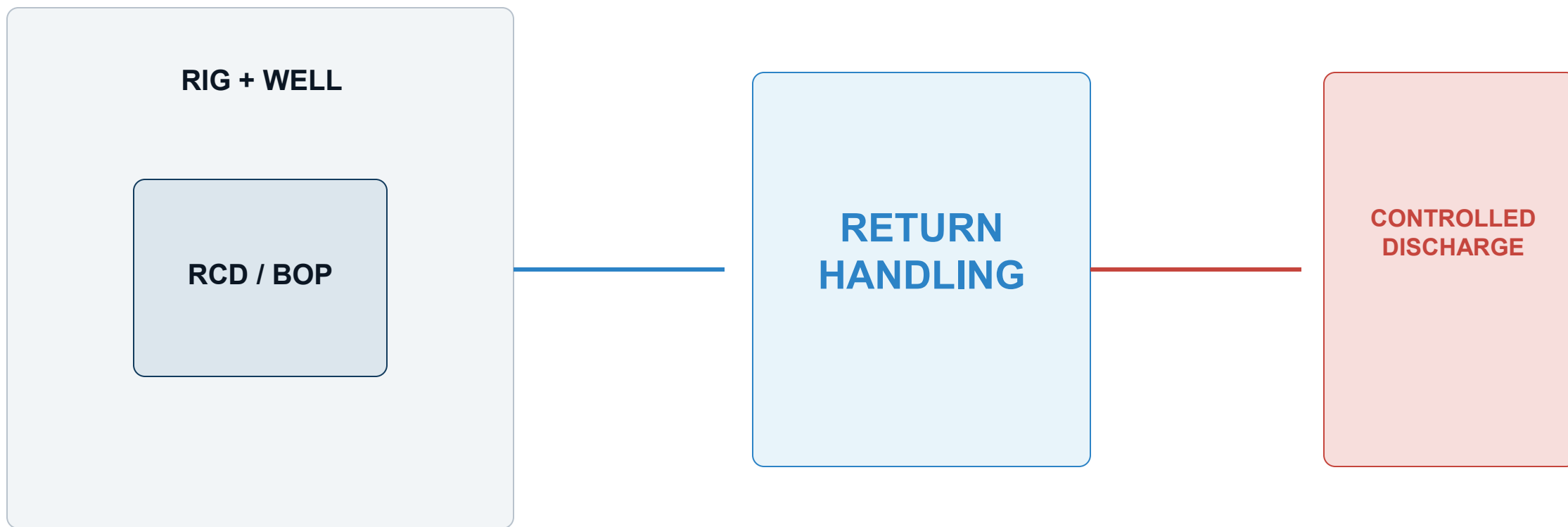
04

Interpretation

- common timestamp
- trend—not snapshot
- alarm ownership
- data quality check

A pressure without its simultaneous gas and liquid rates is incomplete evidence.

Field layout separates people from pressure and discharge hazards



Prevailing wind

Place occupied areas away from high-pressure lines, separator relief paths and hazardous-gas accumulation zones.

Exercise: trace the flow—and find the weak link

The crew reports:

- gas rate is stable
- injection pressure is rising
- standpipe pressure is rising
- surface return pressure is unchanged
- liquid returns are declining

Discuss

1. Which parts of the flow path could create this pattern?
2. What must the driller verify immediately?
3. Which actions could make the situation worse?
4. What would trigger a controlled shutdown?

Use trends and cross-checks—never diagnose from one gauge.

Module 2: equipment limits become operating limits

Flow

Compressor, booster, manifold, drillstring, annulus, choke, separator and discharge.

Pressure

Every component has a rating, and the weakest applicable rating governs.

Information

Synchronized trends connect surface observations to downhole interpretation.

10

minute break

Return ready to connect the equipment to the hydraulics.

Hydraulics and the operating window

Turn geometry, fluid properties and equipment capacity into a controlled range—not a single magic set point.

50 minutes · concepts + worked example + exercise

Bottomhole pressure is a budget of four contributions

Hydrostatic

mixture density × depth

Annular friction

flow through geometry

Surface backpressure

choke / return pressure

Acceleration / transient

gas expansion + change

$$\mathbf{BHP} = \text{hydrostatic} + \text{annular friction} + \text{surface pressure} + \text{transient effects}$$

Gas reduces hydrostatic contribution—but may increase friction and transient sensitivity.

Flow regime changes gas distribution—and liquid holdup

BUBBLY



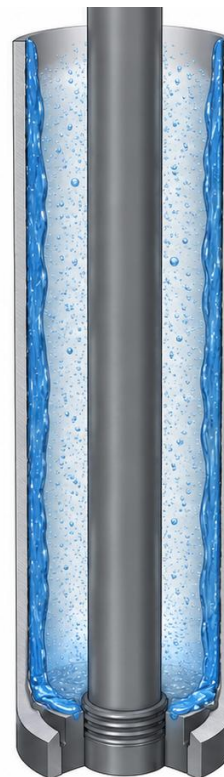
SLUG



CHURN



ANNULAR / MIST



Liquid holdup

Fraction of local pipe volume occupied by liquid at actual pressure, temperature and flow conditions.

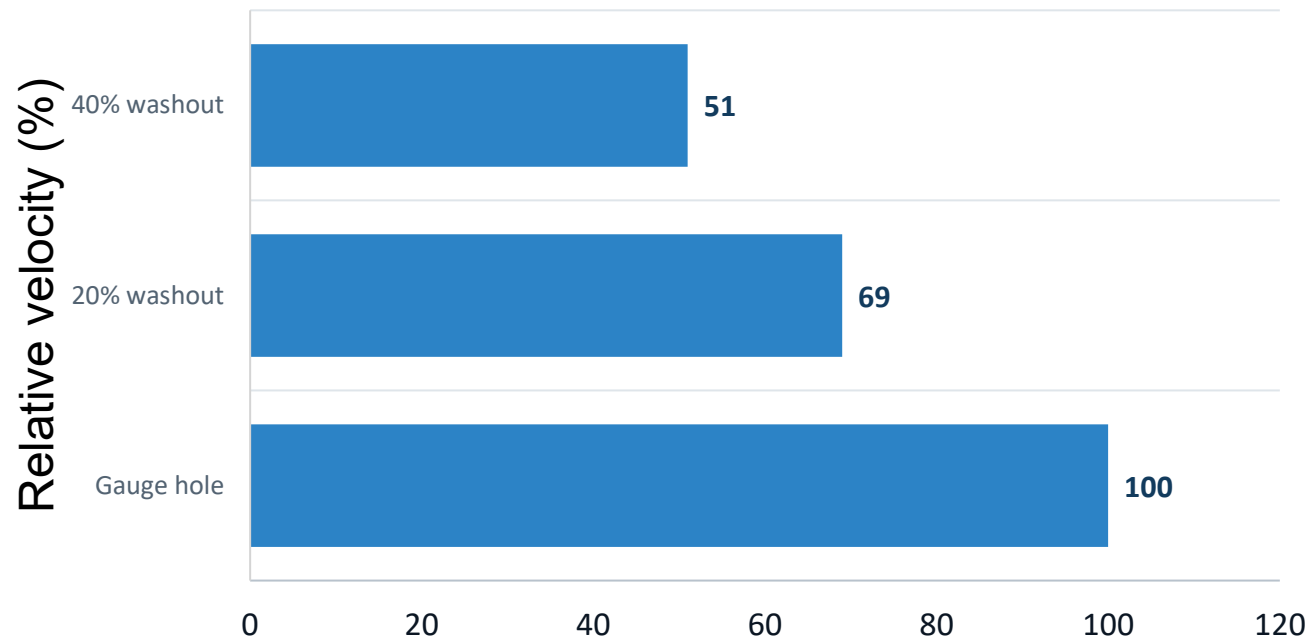
Regime responds to

- gas + liquid rates
- pressure + temperature
- geometry + inclination
- rheology + foam

Field rates are inputs; regime and holdup are downhole states.

The same flow rate produces different velocity in every annulus

Velocity $\approx$ actual volumetric flow / annular area



Illustrative geometry effect; values are normalized.

Why this matters

- washouts create low-velocity pockets
- large collars reduce annular area locally
- gas expands upward, raising velocity
- cuttings beds can form where local margin disappears

Equivalent Liquid Volume translates two-phase flow into a motor check

Plain-language idea

Convert the gas contribution at the tool into an equivalent liquid flow so the combined throughput can be compared with a motor or tool limit.

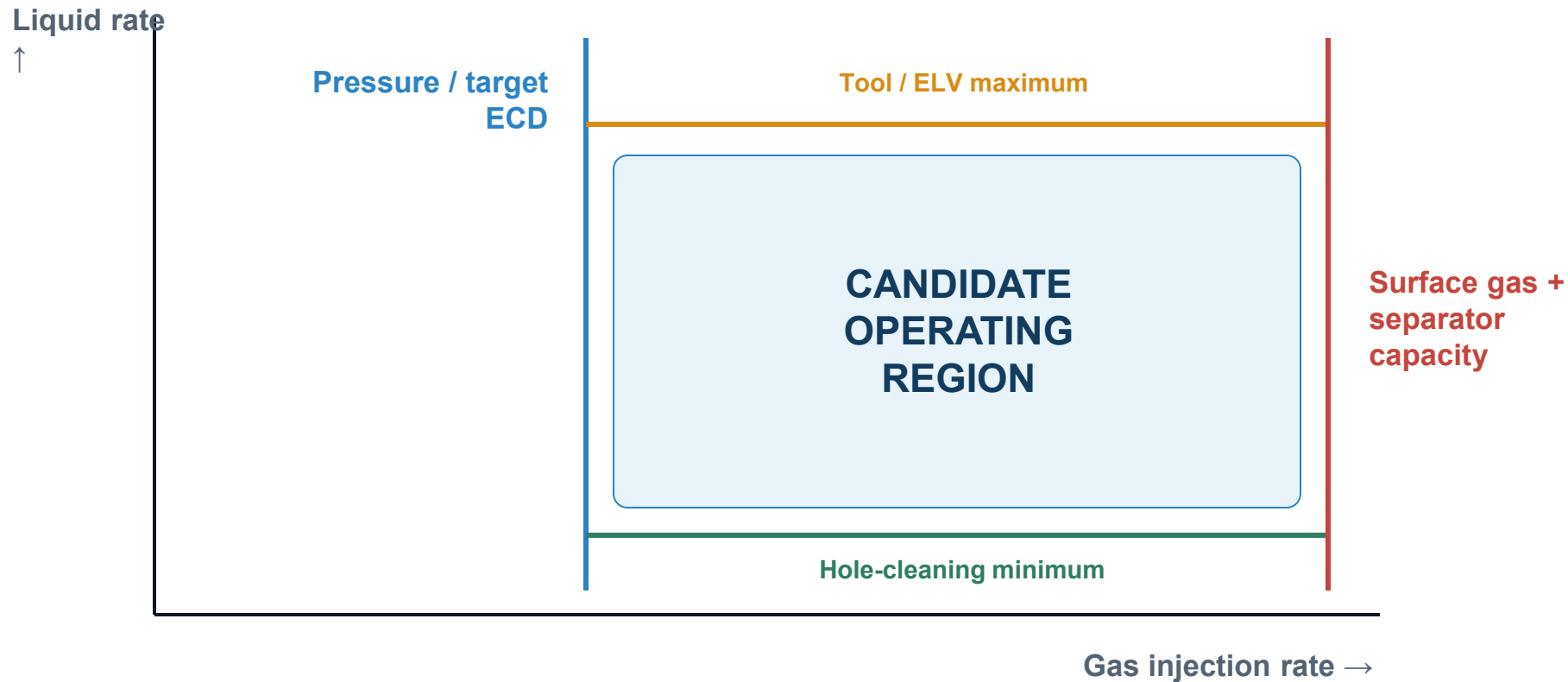
ELV is tool-location specific.

Engineering checks

- gas volume at motor pressure and temperature
- liquid rate through the tool
- motor / MWD maximum and minimum flow
- pressure drop and power-section response
- transient surge during rate changes

Do not add scfm directly to gpm.

Design the operating window in gas–liquid rate space



Build every boundary from the same cases

- same geometry and P–T profile
- same influx / loss assumptions
- same liquid and gas grid
- same equipment ratings

Then shrink the theoretical overlap by uncertainty and operating margin.

Herschel–Bulkley models rheology—not multiphase flow

ROLE / MODEL	CALCULATION IDEA	PROS / USE	LIMITS / WHEN TO ESCALATE
Herschel–Bulkley (rheology submodel)	$\tau = \tau_y + K\dot{\gamma}^n$. Fit τ_y , K and n from rheometer data; calculate apparent viscosity and liquid/foam friction.	Captures yield stress and shear thinning; useful for polymer mud, viscous sweeps and calibrated foam.	Not a multiphase model: no gas expansion, slip, holdup, regime, influx, cuttings or BHP/ECD by itself.
Guo–Ghalambor (aerated hydraulics)	Closed-form four-phase pressure traverse; use liquid properties, then sweep gas rate to BHP/ECD or an operating window.	Purpose-built for aerated drilling; fast, transparent and practical for first-pass rate screening.	Simplified slip/regime/friction. HB enters only if the friction closure is explicitly coupled—do not assume the standard equation does this.
Beggs–Brill / Hagedorn–Brown (holdup cross-checks)	Empirical liquid-holdup and pressure-gradient correlations for gas–liquid production flow.	Quick cross-check on holdup, inclination and pressure trend.	Pipe/tubing basis, no cuttings or bit losses; not primary annular aerated-drilling models.
Mechanistic / transient + field calibration	Solve mass, momentum and energy by regime; may use HB liquid rheology and time-dependent inputs.	Best for slip, regime change, deviation, heat, connections and trips.	Highest data and computation burden; validate with measured pressure and returns.

Use the water level as the formation-pressure datum—not as rock density

Equivalent formation-fluid density

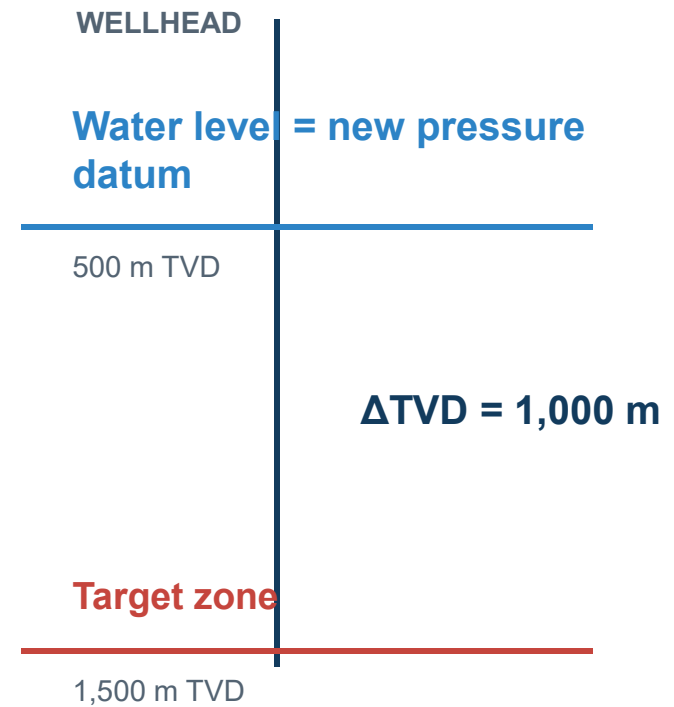
$$MW_{eq,WL_j} = (P_{cf_j} - P_{cWL_j}) \div [0.052 \times (TVD_{cf_j} - TVD_{cWL_j})]$$

Oilfield units: pressure in psi, TVD in ft, result in ppg

$$\text{Metric: } SG_{eq,WL_j} = \Delta P(\text{bar}) \div [0.0980665 \times \Delta TVD(\text{m})]$$

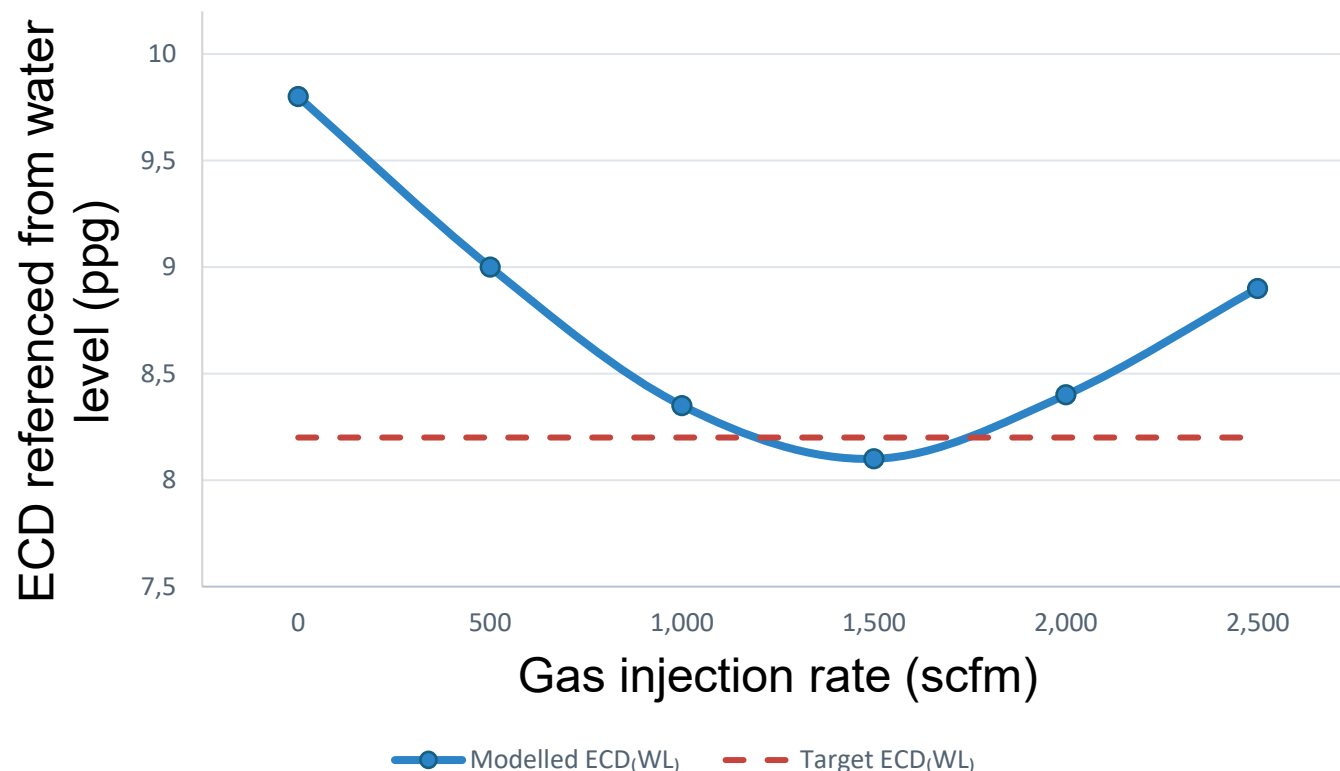
$$P_{cBH,target_j} = P_{cWL_j} + 0.052 \times ECD_{c(target,WL_j)} \times \Delta TVD$$

Use TVD—not measured depth. If the well is artesian, pressurized or not hydrostatic, P_{cWL_j} is not zero; use measured pressure or equivalent head.



Example: $\Delta P = 80 \text{ bar} \rightarrow 0.816 \text{ sg} \approx 6.81 \text{ ppg}$

Why Guo–Ghalambor is the best first-pass aerated-drilling screen



Illustrative ECD sweep. Final rates require field calibration and operating margin.

Preferred for operating-window design

- Purpose-built: injected gas + liquid + formation influx + cuttings.
- Fast: a closed-form pressure traverse enables complete gas-rate sweeps.
- Decision-ready: gives BHP/ECD, void fraction, velocity and density for window screening.
- Transparent: inputs and assumptions remain visible for sensitivity checks.

Best first-pass ≠ universal best. Escalate for strong slip/regime change, foam rheology, severe deviation, rapid transients or reservoir coupling.

Exercise: turn a pressure target into a screened air-rate range

Given

- water level: 500 m TVD
- target zone: 1,500 m TVD
- formation ΔP : 80 bar
- liquid rate: 500 gpm
- air available: 500–2,500 scfm
- desired $ECD_{(WL)}$: 8.2 ppg
- ELV and surface limits apply

Your group must

1. Calculate equivalent formation-fluid density below the water level.
2. Select the model level and list its assumptions.
3. Define the liquid–gas sweep grid.
4. Identify the target-ECD crossing and operating margin.
5. Apply cleaning, ELV, surface and transient limits.
6. State the data required before field approval.

The calculated answer is a candidate range; the drilling program still requires validated geometry, P–T data, equipment curves and site limits.

Module 3: datum, model, sweep, window, validate

Common datum

Reference formation pressure and modelled BHP from the same water-level datum.

Fit-for-purpose model

Use Guo–Ghalambor for screening; escalate when regime or transient physics govern.

Candidate window

Overlap target ECD, cleaning, ELV/tool and surface-capacity boundaries.

Field validation

Step changes, compare trends, recalibrate assumptions and stop outside limits.

The model proposes a range. The operating team earns confidence by validating every step.

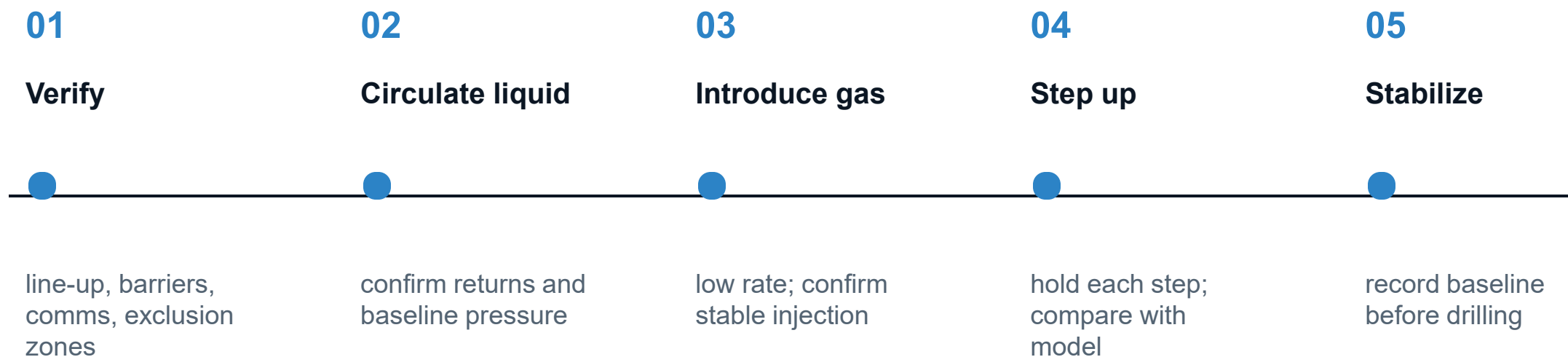
04

Procedures, risk and geothermal application

Translate the operating window into coordinated actions for start-up, drilling, connections and abnormal conditions.

50 minutes · procedures +
scenario

Start-up builds energy in controlled steps



At each hold point: confirm pressure, rates, returns, gas detection and equipment condition.

During drilling, change one control at a time

Primary controls

- liquid rate
- gas rate
- surface backpressure
- ROP / drilling parameters
- chemical injection

Observe response

- injection pressure
- SPP and torque
- return pressure/flow
- cuttings and foam
- pits and gas detectors

Decide

- hold and confirm
- continue step
- return to previous point
- stop drilling
- execute contingency

Simultaneous changes destroy cause-and-effect and can amplify a transient.

Connections are planned transients—not routine pauses

Before

- condition hole
- confirm stable returns
- announce sequence
- set approved rate/backpressure state

During

- manage gas and liquid per procedure
- monitor trapped/backflow pressure
- keep isolation ownership clear
- watch pits and gas alarms

After

- restore liquid first if required
- reintroduce gas in steps
- compare new baseline
- do not resume ROP until stable

Stopping pumps can redistribute gas and liquid; pressure may change after the equipment is quiet.

Hole cleaning must be proven before reducing circulation

Before connection or trip

- reduce or stop ROP
- circulate at validated cleaning condition
- observe cuttings trend—not time alone
- confirm torque/drag and pressure behavior
- account for washout and high-angle sections

During trip

- follow flow-check and barrier procedure
- monitor fill and displacement
- track swab/surge indicators
- manage gas trapped in drillstring
- stop on unexpected flow or losses

A clean-looking shaker is not proof of a clean wellbore.

Abnormal trends require diagnosis before aggressive correction

OBSERVATION	POSSIBLE CAUSES	FIRST CONTROLLED RESPONSE
Injection pressure rises	Restriction, plugging, gas path issue, fluid fallback	Hold/stop step; verify rates and line-up
Return pressure rises	Choke/line restriction, separator overload, foam	Stabilize; check return system and capacity
Returns decline	Losses, measurement error, storage/holdup change	Confirm pits/flow; compare pressures; follow loss plan
Torque/drag increases	Poor cleaning, bed, instability, differential effects	Stop ROP; condition hole within limits
Gas detector alarms	Formation gas, leak, discharge recirculation	Execute site gas response and isolate source

Aerated drilling combines pressure, gas, energy and exposure hazards

High pressure

line failure, hose whip,
stored energy

Hazardous gas

H₂S, CO₂, hydrocarbons,
oxygen deficiency

Ignition

air + hydrocarbons +
ignition source

Discharge

noise, erosion, hot fluid,
solids

Mechanical

rotating equipment, RCD,
pinch points

Chemical

foamer, inhibitor, defoamer
handling

Process upset

separator carryover,
blocked flowline

Human factors

poor handover, unclear
command, fatigue

Controls must be engineered, procedural and practiced—PPE is the final layer.

In geothermal wells, invisible gases can dominate the emergency plan



H_2S

- highly toxic and flammable
- smell cannot be relied upon
- alarms require trained action

CO_2

- colorless and odorless
- can displace oxygen
- can pool in low ground

Use wind direction, gas detection, exclusion zones and the approved escape/rescue plan.

Scenario: losses become inflow as the well heats up

What changes?

- formation pressure response
- return temperature and flashing
- gas composition
- surface gas volume
- separator liquid load

Team decision

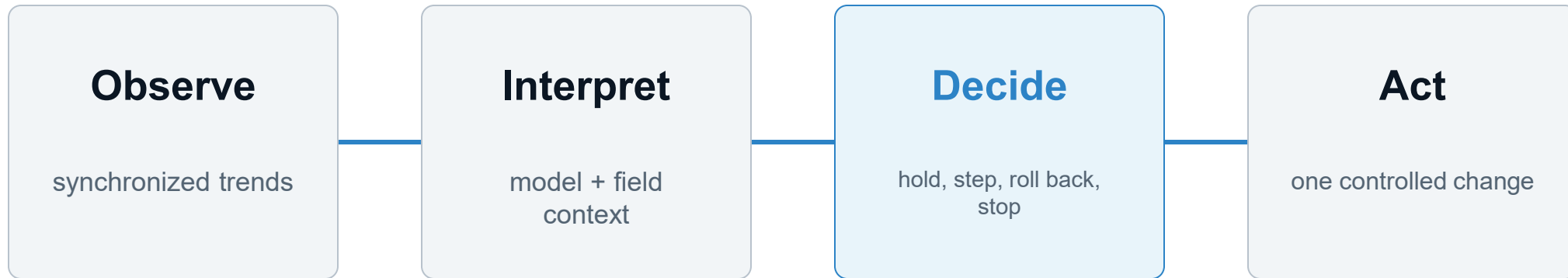
- stabilize rates
- verify flow and pits
- compare BHP trend
- confirm gas alarms
- activate contingency threshold

Communication

- one command channel
- announce each change
- record time and response
- confirm barrier status
- define who can stop work

Group question: which observation would make you stop drilling immediately—and why?

Field control follows a repeatable decision loop



Validate the response, then repeat.

Knowledge check: five questions before we close

1 Why can very high gas rate increase BHP?

2 What four limits define the window?

3 Why is gas expansion a surface issue?

4 What is the safest first response to an abnormal trend?

5 Why can smell never be the H₂S detector?

**Answer in plain language first.
Then add the engineering detail.**

Control the window, not just the rates.

Understand the physics. Trace the system. Respect the limits.
Change deliberately. Validate continuously.

Questions and field application discussion